Airand — Data Model & Data Dictionary

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Summary

Twelve tables hold everything Airand needs to run a zone, bill vendors, pay cyclists and judge viability. One table, **delivery_event**, is the backbone: every state change is an append-only row, so every KPI in the BRD can be computed from facts rather than from counters someone remembered to update.

Design principles

- **Events are the source of truth.** The timestamps on `delivery` (assigned_at, picked_up_at…) are a convenience copy written from events; if they ever disagree, the event wins.
- **Money is a ledger, not a balance.** A wallet's balance is the sum of its `wallet_transaction` rows; holds, settlements and refunds are rows, never edits.
- **Personal data is minimised.** Receiver phone numbers are encrypted at rest and masked in every export; verification codes are stored hashed.
- **Everything is keyed for more zones.** Every operational row carries `zone_id`, so a second zone is data, not a schema change (NFR-13).
- **Configuration lives in rows.** Rates, timers and limits sit on `zone`, so the system admin changes them without a release (FR-50).

What changed from the SRS's nine entities (DR-01 to DR-09). Three tables are added because the requirements need them once the detail is worked out: `wallet_transaction` (the ledger behind DR-03), `shift` (needed to compute cyclist utilisation and pay, FR-45, FR-49), and `admin_user` (the system admin and founder accounts behind role-based access, NFR-09). `payment` keeps one row per delivery summarising what was held, settled and refunded.

All example values in this document are **[Simulated]**.

Entity-relationship diagram

Everything joins to the delivery; every KPI comes from its events



Fork end = many · plain end = one · ? = optional · highlighted = the table everything joins to

Airand ERD · 12 tables, 13 relationships

The delivery sits at the centre: it belongs to one zone and one vendor, is carried by at most one cyclist at a time, has exactly one payment summary, and accumulates an ordered stream of events and any alerts raised about it. Money flows through each vendor's wallet ledger; statements roll a vendor's week up from the same records.

Relationships and rules

Parent	Child	Cardinality	Rule	Why
zone	vendor	1 : N	A vendor belongs to exactly one zone; its pickup pin must lie inside that zone's polygon	RUL-07
zone	cyclist	1 : N	A cyclist is rostered to one zone at a time	BR-02
zone	delivery	1 : N	Copied from the vendor at creation; never changes	Reporting by zone
vendor	wallet	1 : 1	Created at onboarding; never deleted	RUL-03
wallet	wallet_transaction	1 : N	Append-only; balance = sum(amount_ngn)	Ledger principle
vendor	delivery	1 : N	Every delivery has one vendor (the pickup)	BR-05
cyclist	delivery	1 : N (optional)	`delivery.cyclist_id` = the current or final cyclist; null until Assigned; history of reassignments lives in events	FR-15, FR-17
cyclist	shift	1 : N	No overlapping shifts per cyclist	FR-41, FR-45
delivery	delivery_event	1 : N	`seq` is gapless per delivery, starting at 1; rows are never updated or deleted	FR-20, NFR-11
delivery	payment	1 : 1	Created with the request; settled or refunded exactly once	RUL-02, RUL-04, RUL-05

Parent	Child	Cardinality	Rule	Why
payment	wallet_transaction	1 : N (optional)	A payment produces a hold row and, if cancelled, a refund row; top-ups have no payment	FR-06, FR-10
delivery	alert	1 : N (optional)	Alerts can also hang off a cyclist alone (stale location)	FR-28
cyclist	alert	1 : N (optional)	Either `delivery_id` or `cyclist_id` must be set	FR-18
vendor	statement	1 : N	One per vendor per Monday–Sunday week; unique (vendor_id, week_start)	RUL-22
admin_user	delivery_event	1 : N (as actor)	`actor_type` + `actor_id` point to a vendor, cyclist, admin_user or the system; not enforced as a single foreign key	FR-20

Data dictionary — reference and people

Conventions: `uuid` primary keys; `timestamptz` stored in UTC and displayed in West Africa Time; money as whole naira in `integer` (no kobo in the pilot); enums stored as text with a check constraint. PK = primary key, FK = foreign key, NN = not null.

zone — one row per delivery zone; holds its configuration (FR-50)

Column	Type	Keys / null	Description	Example
zone_id	uuid	PK	Zone identifier	—
name	text	NN, unique	Display name	Zone 1 · World Bank Road

Column	Type	Keys / null	Description	Example
polygon	geometry(Polygon)	NN	Boundary used for the in-zone test	2.5 km radius around hub
hub_lat, hub_lng	numeric(9,6)	NN	Hub location	5.4836, 7.0333
open_from, open_to	time	NN	Operating hours (RUL-08)	12:00, 23:30
fixed_rate_ngn	integer	NN, > 0	Fee per delivery (RUL-01)	900
cancel_fee_pct	numeric(4,2)	NN	Fee kept if cancelled after assignment (RUL-04)	0.20
ack_timeout_s, queue_alert_s, handover_grace_s, pickup_alert_s, delivery_alert_s, payment_expiry_s	integer	NN	Timers (RUL-02, 13, 14, 17, 23)	60, 180, 300, 600, 1800, 600
max_weight_kg, max_value_ngn	numeric, integer	NN	Package limits (RUL-24)	8, 50000
created_at, updated_at	timestamptz	NN	Audit	—

vendor — a business that requests deliveries

Column	Type	Keys / null	Description	Example
vendor_id	uuid	PK	—	—
zone_id	uuid	FK → zone, NN	Home zone	—
name	text	NN	Trading name	Mama Chi's Kitchen

Column	Type	Keys / null	Description	Example
contact_name	text	NN	Person responsible	Chioma
phone	text	NN, unique	E.164 format	+2348035551234
pickup_address	text	NN	As written	Plot 22, World Bank Road
pickup_lat, pickup_lng	numeric(9,6)	NN	Pinned at onboarding	—
category	text	NN	food · retail · print · other	food
trial_start	date	NN	Start of 7-day free trial (RUL-06)	2026-10-01
status	text	NN	active · paused · offboarded	active
onboarded_by	uuid	FK → admin_user	Who onboarded them (FR-37)	—
created_at	timestamptz	NN	—	—

wallet and **wallet_transaction** — the vendor's prepaid balance as a ledger

Column	Type	Keys / null	Description	Example
wallet.wallet_id	uuid	PK	—	—
wallet.vendor_id	uuid	FK → vendor, NN, unique	One wallet per vendor	—
wallet.account_number	text	NN, unique	Virtual account from the payment provider (PayHold, demo)	9012448317

Column	Type	Keys / null	Description	Example
wallet_transaction.txn_id	uuid	PK	—	—
wallet_transaction.wallet_id	uuid	FK → wallet, NN	—	—
wallet_transaction.payment_id	uuid	FK → payment, null	Set for hold and refund rows	—
wallet_transaction.type	text	NN	topup · hold · refund · adjustment	hold
wallet_transaction.amount_ngn	integer	NN, ≠ 0	Signed: + top-up/refund, − hold	−900
wallet_transaction.provider_ref	text	null	Provider reference for top-ups	PH-77A31
wallet_transaction.created_at	timestamptz	NN	—	—

cyclist — an Airand rider

Column	Type	Keys / null	Description	Example
cyclist_id	uuid	PK	—	—
zone_id	uuid	FK → zone, NN	Current zone	—
name, phone	text	NN	—	Kelechi D.

Column	Type	Keys / null	Description	Example
state	text	NN	off_shift · available · assigned · to_vendor · at_vendor · delivering (RUL-12)	available
last_lat, last_lng, last_seen_at	numeric, timestamptz	null	Last reported location; stale after 5 min (FR-18)	—
retainer_ngn	integer	NN	Daily shift retainer (RUL-10)	2310
bonus_threshold, bonus_per_delivery_ngn	integer	NN	Deliveries before bonus; bonus per extra delivery	6, 150
trained_at	date	null	Training record (BR-20)	2026-09-28
status	text	NN	active · suspended · left	active

shift — a rostered and actual working period

Column	Type	Keys / null	Description	Example
shift_id	uuid	PK	—	—
cyclist_id, zone_id	uuid	FK, NN	—	—
planned_start, planned_end	timestamptz	NN	From the weekly roster (RUL-09)	15:30–23:30
actual_start, actual_end	timestamptz	null	From Start/End shift in the app (FR-41)	15:34–23:31

admin_user — the system admin and the founder

Column	Type	Keys / null	Description	Example
admin_id	uuid	PK	—	—
zone_id	uuid	FK → zone, null	Null = all zones	—
name	text	NN	—	Chinedu
role	text	NN	system_admin · founder (NFR-09)	system_admin
phone, email	text	NN	—	—

Data dictionary — deliveries, money and records

delivery — one row per request; the hub of the model

Column	Type	Keys / null	Description	Example
delivery_id	uuid	PK	—	—
ref	text	NN, unique	Human reference shown on screens	AR-1042
zone_id, vendor_id	uuid	FK, NN	—	—
cyclist_id	uuid	FK → cyclist, null	Current or final cyclist	—
raised_by	text	NN	vendor · admin (phone-in, FR-12)	vendor
package_category	text	NN	food · documents · parcel · other (FR-52)	food

Column	Type	Keys / null	Description	Example
limits_confirmed	boolean	NN, must be true	Vendor ticked the limits box	true
dropoff_address	text	NN	As entered or pasted	14 Okigwe Close, Housing Area H
dropoff_lat, dropoff_lng	numeric(9,6)	NN	Geocoded pin, inside zone	—
receiver_phone_enc	bytea	NN	Encrypted; masked in exports (NFR-08)	+234 803 ··· 4417
note	text(140)	null	For the cyclist	Blue gate
fee_ngn	integer	NN	Rate at time of request; 0 in trial	900
state	text	NN	Current lifecycle state (11 values, FR-19)	picked_up
code_hash	text	null	Hash of the 4-digit code; set on assignment	—
tracking_token	text	NN, unique	≥ 128-bit, for the receiver link (NFR-10)	—
created_at, paid_at, activated_at, assigned_at, arrived_at, picked_up_at, delivered_at, closed_at	timestamptz	null	Convenience copies of event times	—

Column	Type	Keys / null	Description	Example
vendor_delay	boolean	NN, default false	Handover > grace after arrival (FR-23)	false
without_code	boolean	NN, default false	Closed by phone check (FR-25)	false
outcome	text	null	delivered · failed · cancelled · expired	delivered
outcome_reason	text	null	From fixed list	receiver_unreachable

delivery_event — append-only history; the source for every KPI

Column	Type	Keys / null	Description	Example
event_id	uuid	PK	—	—
delivery_id	uuid	FK → delivery, NN	—	—
seq	integer	NN; unique (delivery_id, seq)	Order within the delivery	6
event_type	text	NN	See event taxonomy	PICKED_UP
from_state, to_state	text	null	Null when the event does not change state	assigned → picked_up
actor_type	text	NN	vendor · cyclist · admin · system	cyclist
actor_id	uuid	null	Null for system	—
server_ts	timestamptz	NN	Authoritative time	2026-10-09 20:18:04 UTC

Column	Type	Keys / null	Description	Example
device_ts	timestamptz	null	Time on the phone, for offline actions (FR-44)	—
lat, lng	numeric(9,6)	null	Where it happened, if known	—
reason, note	text	null	Mandatory for overrides and failures	—

payment — one money summary per delivery

Column	Type	Keys / null	Description	Example
payment_id	uuid	PK	—	—
delivery_id	uuid	FK, NN, unique	—	—
method	text	NN	wallet · transfer	wallet
amount_held_ngn	integer	NN	Fee held at request	900
amount_settled_ngn	integer	NN, default 0	Recognised as Airand revenue	900
amount_refunded_ngn	integer	NN, default 0	Returned to wallet	0
cancel_fee_ngn	integer	NN, default 0	Kept on post-assignment cancellation	0
status	text	NN	pending · held · settled · refunded · part_refunded · expired	settled
held_at, settled_at	timestamptz	null	—	—
		check	held = settled + refunded once final	—

alert — something the system admin must look at

Column	Type	Keys / null	Description	Example
alert_id	uuid	PK	—	—
zone_id	uuid	FK, NN	—	—
delivery_id, cyclist_id	uuid	FK, null; at least one set	Subject	AR-1042
type	text	NN	ack_timeout · queued · not_picked_up · late_delivery · vendor_delay · code_locked · stale_location · sync_delay	not_picked_up
raised_at	timestamptz	NN	—	—
resolved_at	timestamptz	null	—	—
resolved_by	uuid	FK → admin_user, null	—	Chinedu
action	text	null	What was done	reassigned

statement — a vendor's week

Column	Type	Keys / null	Description	Example
statement_id	uuid	PK	—	—
vendor_id	uuid	FK, NN	—	—
week_start, week_end	date	NN; unique (vendor_id, week_start)	Monday–Sunday	2026-09-28, 2026-10-04
delivered, failed, cancelled	integer	NN	Counts	46, 1, 2

Column	Type	Keys / null	Description	Example
fees_ngn, cancel_fees_ngn, topups_ngn	integer	NN	Totals	42300, 180, 45000
opening_ngn, closing_ngn	integer	NN	Wallet balance at start and end	4780, 7300
vendor_delays	integer	NN	Count for the coaching note	3
generated_at	timestamptz	NN	Monday 06:00	—

Event taxonomy

Twenty-two event types cover the whole lifecycle in the Future-State Design. Events that change state carry `from_state` and `to_state`; the rest record something that happened without moving the delivery. The Feeds column names the measures each event supports; the KPI definitions in the next step are written against these event types.

Event type	From → to	Actor	Written when	Feeds
REQUEST_CREATED	— → pending_payment	vendor / admin	Request submitted	Demand by hour
PAYMENT_HELD	pending_payment → active	system	Fee held from wallet or transfer confirmed	Payment latency
PAYMENT_EXPIRED	pending_payment → expired	system	10 min without payment	Expiry rate
QUEUED	active → awaiting_cyclist	system	No cyclist available	Queue rate, queue time
CYCLIST_PROMPTED	—	system	A cyclist is offered the job (one per attempt)	Offers per assignment
CYCLIST_ACCEPTED	active / awaiting / reassigning → assigned	cyclist	Accept tapped	Request-to-assignment

Event type	From → to	Actor	Written when	Feeds
ACK_TIMEOUT	— (offer withdrawn, next cyclist prompted)	system	60 s without accept	Unresponsive rate per cyclist
REASSIGNED	assigned → reassigning	admin	Admin override with reason	Override rate
ARRIVED_AT_VENDOR	—	cyclist / system (geofence)	Tap or 50 m for 30 s	Cyclist pickup wait
VENDOR_DELAY_FLAGGED	—	system	No pickup 5 min after arrival	Vendor delay rate
PICKED_UP	assigned → picked_up	cyclist	Handover confirmed	Vendor handover time
REFUSED_AT_PICKUP	assigned → cancelled	cyclist	Package outside limits (FR-53)	Refusal rate
ARRIVED_AT_DROPOFF	—	cyclist	Slide at the door	Last-metre wait
CODE_FAILED	—	cyclist	Wrong code entered	Code friction
CODE_LOCKED	—	system	Third wrong code	Alerts
DELIVERED	picked_up → delivered	cyclist	Correct code	Delivery time, on-time %
DELIVERED_WITHOUT_CODE	picked_up → delivered	cyclist	Phone-confirmed handover	Review volume
FAILED	picked_up → failed	cyclist / admin	Receiver unreachable or refused (RUL-19)	Failure rate and reasons
CANCELLED	active / awaiting / assigned → cancelled	vendor / admin	Cancel with reason	Cancellation rate, fees kept
CLOSED	delivered / failed → closed	admin / system	Review done or 24 h after delivery	Close lag

Event type	From → to	Actor	Written when	Feeds
MESSAGE_SENT	—	system	Code and link sent; channel recorded	Message cost, SMS fallback rate
SYNC_LATE	—	system	Offline action arrived > 2 min late	Connectivity issues

Rules. `seq` increments by one per delivery. A state-changing event is rejected if `from_state` does not equal the delivery's current state, which is how the database enforces the lifecycle. Offline actions keep `device_ts` but are ordered by `server_ts`.

Data quality, privacy and retention

Area	Rule	Enforced by	Requirement
Lifecycle integrity	A state change must start from the delivery's current state; `seq` is gapless	Database check on insert into delivery_event	FR-19, FR-20
Immutability	delivery_event and wallet_transaction rows are never updated or deleted; corrections are new rows	No update/delete grants on those tables	NFR-11
Money balances	payment: held = settled + refunded once final; wallet balance never below 0	Check constraint; hold refused if insufficient	RUL-02, RUL-04
Location sanity	Pins must fall inside Nigeria's bounding box; drop-offs inside the zone polygon	Geocoding step and zone test	FR-03
Time	Server time is authoritative; device time is kept but never used for ordering	Event writer	FR-44

Area	Rule	Enforced by	Requirement
Receiver phone	Encrypted at rest; visible only to the assigned cyclist during the job, the system admin and the requesting vendor; masked in reports and exports	Column encryption + role-based views	NFR-08
Codes and links	Codes stored hashed; tracking token stops resolving 24 h after the delivery ends	Application	NFR-10, FR-40
Retention	All operational records kept ⩾ 12 months, exportable as CSV for CLRD inspection and disputes; receiver phone purged after 12 months	Scheduled job	BR-22, NFR-12
Synthetic data	Every dataset made for the portfolio is labelled [Simulated] in its file name and first row; no real phone numbers are used	Generator script	Portfolio rule

Traceability and next steps

SRS data requirement	Table(s) here
DR-01 Zone	zone
DR-02 Vendor	vendor
DR-03 Wallet	wallet, wallet_transaction
DR-04 Cyclist	cyclist, shift
DR-05 Delivery	delivery
DR-06 Event	delivery_event
DR-07 Payment	payment

SRS data requirement	Table(s) here
DR-08 Statement	statement
DR-09 Alert	alert
NFR-09 role-based access	admin_user

Next in Phase 8 (data track)

- [ ] KPI definitions: each BRD success criterion and management measure written once, as a formula over these tables and events
- [ ] Synthetic dataset: 90 days, 8 vendors (2 anchors), 10 cyclists, demand ramping from ~30 to ~70 deliveries a day with a 7–11 PM peak, labelled [Simulated]
- [ ] SQL for every KPI against the dataset
- [ ] Dashboard answering the founder's question: is Zone 1 viable, and what should change?